

IK 518: Ancient Indian Materials and Chemistry

Course Code: IK 518

Course Name: Ancient Indian Materials and Chemistry

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: UG/PG/PhD

Mutual Exclusion: None

Approval: 61st BoA

Preamble

This course interlinks the science of materials in ancient India to modern knowledge. It explores how early practices in metallurgy, chemistry, medicine, and engineering laid the foundation for today's materials science. The course emphasizes both historical depth and modern interpretation, showing how ancient insights continue to inspire sustainable and innovative approaches in science and technology. This course brings together UG and PG students from engineering, materials science, chemistry, and related disciplines, along with the students of IKSMHA centre. Care has been taken to present the materials in a gradual manner to instill confidence in the minds of the students.

Course Modules with Quantitative Lecture Hours

- **Unit 1: Foundations of Ancient Indian Science:** Basic principles of scientific thought and technological practice in ancient India, emphasizing how material innovations, experimental techniques, and design philosophies shaped early engineering and craftsmanship. Overview of key materials such as iron, ceramics, textiles, and construction practices that evolved from empirical observation and indigenous knowledge systems. Reflection on how these traditions resonate with contemporary materials science, sustainable design, and circular-economy approaches. A brief comparison with Greek and Roman materials and technologies will provide a broader civilizational perspective. (6 hours)
- **Unit 2: History of ancient Chemistry and metallurgy:** Alchemical traditions and their connections to metallurgy, medicine, and philosophy. Chemical operations described in classical texts: distillation, sublimation, calcination, incineration, and alloying. Early theories of matter, including *Panchamahabhutas* and their analogies to chemical elements. Rasashastra growth as an experimental science, developing formulations for ailments such as, respiratory, skin, musculoskeletal etc. Mining practices and mineral extraction. Major metals and alloys in ancient India. Iron in the Vedic period and the innovation of Wootz steel. Zinc extraction and distillation techniques unique to India. (14 hours)
- **Unit 3: Glass, Ceramics, and Organic Materials:** Development of glass technologies, highlighting how artisans mastered high-temperature reactions, flux control, and coloring techniques. Ceramic traditions from Harappa to later periods, including Harappan bricks, mortars, and concretes, with emphasis on optimization of clay composition and firing conditions. Pigments, dyes, and mordants in textile chemistry, including natural dyes, resins, gums, and textiles from plant and animal sources. Organic-inorganic composites in arts and crafts. Scientific analysis of artifacts and approaches to preservation. Selected theoretical approaches for studying ancient materials and technologies. The unit links ancient synthesis, thermal processing, and surface chemistry with modern materials science concepts such as vitrification, sintering, and polymer chemistry. (8 hours)

- **Unit 4: Environment, Sustainability, and Modern Relevance:** Ancient engineering practices in the context of sustainability, including stone processing, plastering, and waterproofing techniques. Principles of green materials and circular economy as reflected in traditional construction and craft practices. Environmental consciousness in ancient India, applications of modern tools such as spectroscopy, and surface analysis to study ancient artifacts. Interpreting traditional knowledge through the lens of materials science. Relevance to nanoscience, coatings, biomedicine, and sustainable technologies. Brief comparative insights from other ancient civilizations will be added to highlight shared sustainable practices. (14 hours)

Laboratory/Practical/Tutorial Modules

The course will involve assignments.

Text Books

1. Ray, P. C. (1902–1909). *A history of Hindu chemistry: From the earliest times to the middle of the sixteenth century A.D.* (Vols. 1–2). Calcutta: The Bengal Chemical and Pharmaceutical Works.
2. Chattopadhyaya, D. (1986–1991). *History of science and technology in ancient India* (Vols. 1–3). Calcutta: Firma KLM.
3. Tripathi, V. (2008). *History of iron technology in India (from the beginning to pre-modern times)*. Delhi: Rupa & Co.
4. Agrawal, D. P. (2000). *Ancient metal technology and archaeology of South Asia: A pan-Asian perspective*. New Delhi: Aryan Books International.

References

1. Narain, S., & Chandra, N. (2023). *Archaeometallurgy of India from Rigveda to Vālmīki Rāmāyaṇa*. Roorkee: AG Tech Publishing.
2. Divakar, R., Murty, S. V. S. N., Srikanth, S., & Gokhale, A. A. (2023). *Indian Metallurgy: The Platinum Years*. Singapore: Springer.
3. Biswas, A. K. (2005). *Science in Archaeology and Archaeo-materials*. D.K. Printworld Pvt. Ltd.